H

NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
Higher 2

CANDIDATE NAME	**SUGGESTED SOLUTION**		
CLASS		TUTOR'S NAME	
CENTRE NUMBER	S	INDEX NUMBER	

PHYSICS 9478/01

Paper 1 Multiple Choice **23 September 2026**

1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, glue or correction fluid.
Write your name, class, Centre number and index number in the spaces at the top of this page.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
The use of an approved scientific calculator is expected, where appropriate.

This document consists of **22** printed pages.

Data

speed of light in free space	c	=	3.00×10^{8} m s^{-1}
permeability of free space	μ_0	=	$4\pi \times 10^{-7}$ H m^{-1}
permittivity of free space	ε_0	=	8.85×10^{-12} F m^{-1}
	$(\frac{1}{4\pi\varepsilon_0}$	=	8.99×10^{9} m F^{-1})
elementary charge	e	=	1.60×10^{-19} C
the Planck constant	h	=	6.63×10^{-34} J s
unified atomic mass constant	u	=	1.66×10^{-27} kg
rest mass of electron	m_e	=	9.11×10^{-31} kg
rest mass of proton	m_p	=	1.67×10^{-27} kg
molar gas constant	R	=	8.31 J K^{-1} mol^{-1}
Avogadro constant	N_A	=	6.02×10^{23} mol^{-1}
Boltzmann constant	k	=	1.38×10^{-23} J K^{-1}
gravitational constant	G	=	6.67×10^{-11} N m^2 kg^{-2}
acceleration of free fall	g	=	9.81 m s^{-2}

Formulae

uniformly accelerated motion	s	=	$ut + \frac{1}{2}at^2$
	v^2	=	$u^2 + 2as$
work done on / by gas	W	=	$p\,\Delta V$
pressure	p	=	$\frac{F}{A}$
gravitational potential	ϕ	=	$-\frac{GM}{r}$
temperature	T / K	=	$T / °C + 273.15$
pressure of an ideal gas	p	=	$\frac{1}{3}\frac{Nm}{V} < c^2 >$
mean translational kinetic energy of an ideal gas particle	E	=	$\frac{3}{2}kT$
displacement of particle in s.h.m.	x	=	$x_0 \sin \omega t$
velocity of particle in s.h.m.	v	=	$v_0 \cos \omega t = \pm\omega\sqrt{\left(x_0{}^2 - x^2\right)}$
electric current	I	=	$Anvq$
resistors in series	R	=	$R_1 + R_2 + \ldots$
resistors in parallel	$1/R$	=	$1/R_1 + 1/R_2 + \ldots$
capacitors in series	$1/C$	=	$1/C_1 + 1/C_2 + \ldots$
capacitors in parallel	C	=	$C_1 + C_2 + \ldots$
energy in a capacitor	U	=	$\frac{1}{2}QV = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}CV^2$
charging a capacitor	Q	=	$Q_0\left[1 - e^{-\frac{t}{\tau}}\right]$
discharging a capacitor	Q	=	$Q_0 e^{-\frac{t}{\tau}}$
RC time constant	τ	=	RC
electric potential	V	=	$\frac{Q}{4\pi\varepsilon_0 r}$
alternating current / voltage	x	=	$x_0 \sin \omega t$
magnetic flux density due to a long straight wire	B	=	$\frac{\mu_0 I}{2\pi d}$
magnetic flux density due to a flat circular coil	B	=	$\frac{\mu_0 NI}{2r}$
magnetic flux density due to a long solenoid	B	=	$\mu_0 nI$
energy states for quantum particle in a box	E_n	=	$\frac{h^2}{8mL^2}n^2$
radioactive decay	x	=	$x_0 e^{-\lambda t}$
radioactive decay constant	λ	=	$\frac{\ln 2}{t_{\frac{1}{2}}}$

1 The kinetic energy E of an object rotating about its centre of mass is calculated using

$$E = \frac{1}{2}k\omega^2$$

where ω is the angular speed of rotation, and k is a constant.

What is the unit for k?

A kg m^2 **B** kg m s^{-1} **C** kg m s^{-2} **D** kg m^2 s^{-4}

Ans: **A**

$k = \frac{2E}{\omega^2}$ → [k] = (kg m^2 s^{-2}) (rad s^{-1})$^{-2}$ = kg m^2

2 The following measurements were taken of a uniform metal disc.

mass m = (265.2 ± 0.5) g

diameter D = (10.3 ± 0.1) cm

thickness t = (1.36 ± 0.02) cm

electrical resistance R = (20 ± 1) Ω

The density ρ of the disc is determined using the above measurements. Which measurement contributes the most to the uncertainty of the value of ρ?

A m **B** D **C** t **D** R

Ans: **B**

$\rho = \frac{m}{\left(\frac{\pi D^2}{4} \times t\right)}$ → $\pm\frac{\Delta\rho}{\rho} = \pm\left(\frac{\Delta m}{m} + 2\frac{\Delta D}{D} + \frac{\Delta t}{t}\right) = \pm(0.19\% + 2\times 0.97\% + 1.5\%)$

3 A spring obeying Hooke's law has an unstretched length of 45 mm and a spring constant of 380 N m^{-1}.

What is the increase in the elastic potential energy of the spring when its length increases from 50 mm to 60 mm?

A 0.019 J **B** 0.038 J **C** 0.076 J **D** 0.21 J

Ans: **B**

Extension goes from 5 mm to 15 mm.

$$\text{increase in EPE} = \frac{1}{2}k\left(x_f^2 - x_i^2\right) = \frac{1}{2}(380)\left(0.015^2 - 0.005^2\right)$$
$$= 0.038 \text{ J}$$

A: use (1/2)k(Δx)^2 ; C: forgot × ½ ; D: took length instead of extension

4 A rigid body is acted upon by a number of coplanar forces. Which of the following statements is correct?

A When the vector sum of all the forces acting on the body is zero, the object must be in a state of static equilibrium.

B When the coplanar forces are a couple, the magnitude of the torque of the couple on the object does not depend on the choice of the pivot point.

C When acted upon by three non-parallel coplanar forces whose lines of action intersect at a single point, the body must be in static equilibrium.

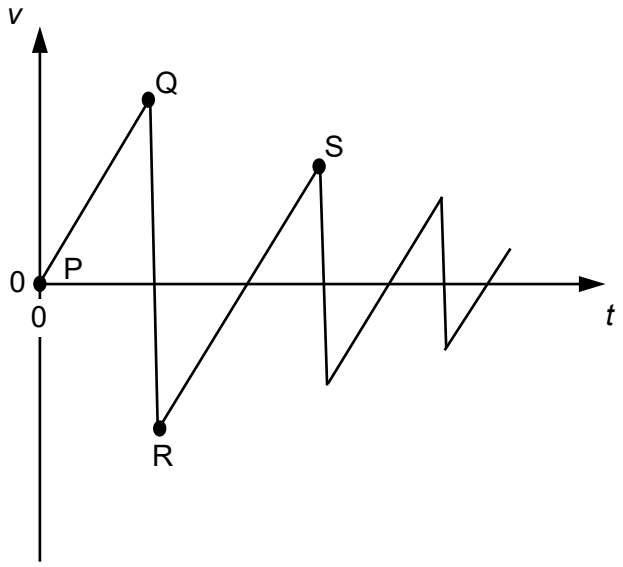
D When acted upon by two forces of equal magnitude and opposite directions, the body will always experience a net turning effect.

Ans: **B**

A couple creates a pure turning effect. Because the two constituent forces are equal in magnitude and opposite in direction, the distance variables to any arbitrary origin cancel out algebraically when you sum their moments. This makes the torque invariant across any reference point in space.

5 A ball is held above the ground and released. It falls to the ground and bounces several times.

The graph shows the variation with time t of the velocity v of the ball.



Four points on the graph are labelled P, Q, R and S.

Which statement is **not** correct?

A The area under line PQ represents the initial height of the ball.

B The collisions of the ball with the ground are inelastic.

C The gradient of the line RS represents the acceleration due to free fall.

D The maximum upwards velocity of the ball is reached at point Q.

Ans: **D**

Option A: since point P represents the point of impact with the ground, the displacement from P to Q represents the initial height it was dropped from.

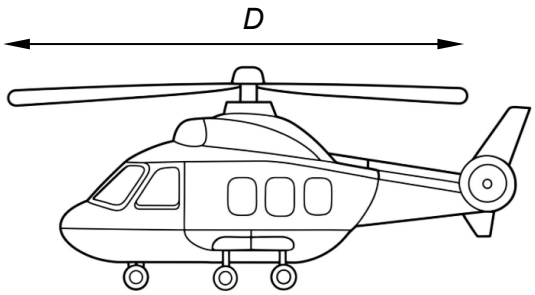
Option B: Seeing that the maximum velocity between subsequent bounces gets smaller and smaller, this means that there is KE loss in the collisions.

Option C: all the positively sloped gradients represent acceleration due to free fall.

Option D: line PQ represents the ball falling to the ground, implying that the sign convention for the positive direction is downwards. Therefore, Q is the maximum downward velocity.

6 A helicopter hovers stationary in mid-air. Its rotor blades sweep out a circular path of diameter D and push down a continuous column of air of density ρ at a uniform speed v. The air above the blades is taken to be stationary.

Which expression correctly represents the average power P delivered to the air column by the blades?



A $\frac{1}{8}\pi\rho D^2v^3$ **B** $\frac{1}{2}\pi\rho D^2v^3$ **C** $\frac{1}{4}\pi\rho D^2v^3$ **D** $\frac{1}{4}\pi\rho D^2v^2$

Ans: **A**

Mass flow rate:

$$A = \pi r^2 = \frac{\pi D^2}{4}$$

$$\frac{\Delta m}{\Delta t} = \rho A v = \rho\left(\frac{\pi D^2}{4}\right)v \Rightarrow F = \frac{\Delta m}{\Delta t}v = \frac{1}{4}\rho\pi D^2 v^2 \quad \text{(not needed)}$$

$$P = \frac{\text{change in KE}}{\text{time taken}} = \frac{1}{2}\left(\frac{\Delta m}{\Delta t}\right)v^2 = \frac{1}{2}\left[\rho\left(\frac{\pi D^2}{4}\right)v\right]v^2$$

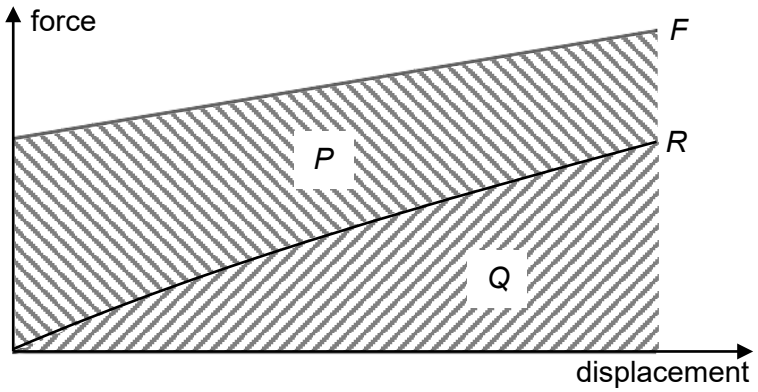
$$= \frac{1}{8}\pi\rho D^2 v^3$$

B: forgot to divide diameter

C: Used $P = Fv$ directly. (forgot factor of ½ in KE formula)

D: Used $P = F$

7 A body is accelerated from rest by a horizontal force F on a smooth surface through air. The magnitudes of F and air resistance R vary with displacement as shown in the graph below.



Which area represents the thermal energy dissipated?

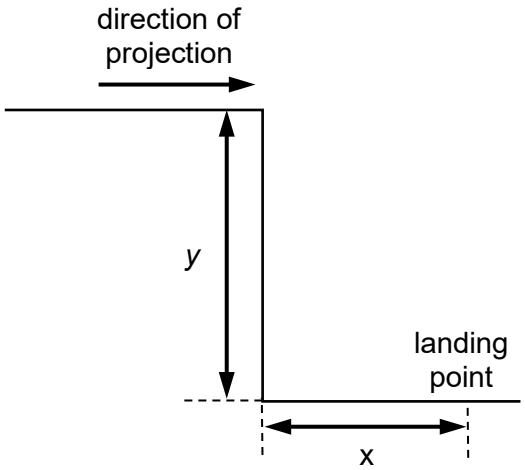
A P **B** Q **C** $P - Q$ **D** $P + Q$

Ans: **B**

Thermal energy dissipated = Work done against air resistance R

= Area under force-displacement graph for R

8 An object is projected horizontally. The object falls a vertical distance y and travels a horizontal distance x before landing on the ground.



A second object is projected horizontally with the same initial velocity and falls a vertical distance $4y$ before landing on the ground.

Assume that air resistance is negligible.

Which horizontal distance does the second object travel?

A x **B** $2x$ **C** $4x$ **D** $16x$

Ans: **B**

First object,

$$s_y = u_y t + \frac{1}{2} a_y t^2$$

$$y = \frac{1}{2} g t_1^2$$

$$t_1 = \sqrt{\frac{2y}{g}}$$

Second object,

$$s_y = u_y t + \frac{1}{2} a_y t^2$$

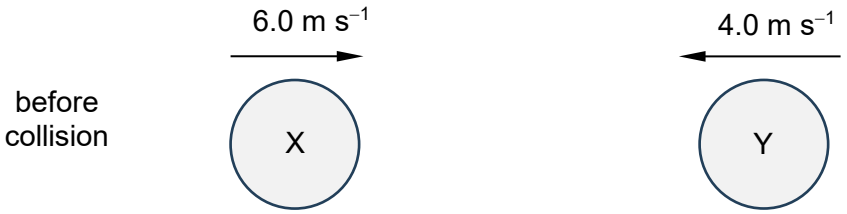
$$4y = \frac{1}{2} g t_2^2$$

$$t_2 = \sqrt{\frac{2(4y)}{g}} = 2\sqrt{\frac{2y}{g}} = 2t_1$$

With constant horizontal velocity, twice the time of travel means twice the horizontal distance, $2x$.

9 Body X of mass 2.5 kg undergoes a head-on collision with body Y of mass 1.0 kg. Bodies X and Y have initial speeds of 6.0 m s^{-1} and 4.0 m s^{-1} respectively.

The two bodies are in contact for a time of 20 ms. After the collision, body X continues in the same direction with speed 2.0 m s^{-1}.



What is the magnitude of the average force exerted on body Y during this collision?

A 10 N **B** 200 N **C** 250 N **D** 500 N

Ans: **D**

Change in momentum of X:

$$\Delta p_x = m_x(v_x - u_x) = (2.5)(2.0 - 6.0) = -10.0 \text{ N s}$$

By N3L:

$$\Delta p_y = -\Delta p_x = 10.0 \text{ N s}$$

$$F_{avg} \Delta t = \Delta p_y = 10.0 \text{ N s}$$

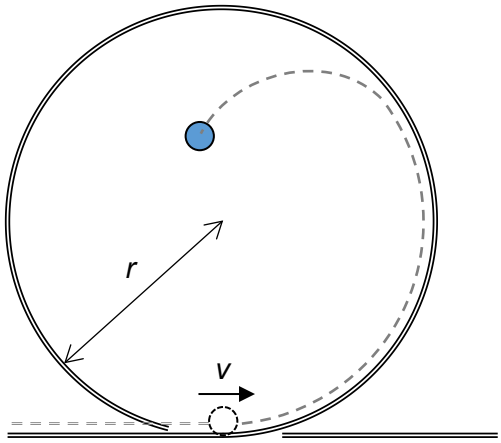
$$F_{avg} = \frac{10}{20 \times 10^{-3}} = 500 \text{ N}$$

A: forgot to divide by time

B: used mass of Y

C: didn't calculate change in momentum, used final momentum of Y

10 A body enters the vertical circular track of radius r shown below with speed v.



If the body falls off the track as shown above, what is a possible value of v?

A $\sqrt{gr}$ **B** $\sqrt{2gr}$ **C** $\sqrt{4gr}$ **D** $\sqrt{6gr}$

Ans: **C**

Min speed at mid-height = 0

Min v to reach mid-height = $\sqrt{2gr}$ since $\frac{1}{2}mv^2 - 0 = mgr$ (conservation of energy)

Min speed at top = $\sqrt{gr}$ since $mg < m\frac{v^2}{r}$ (N2L)

Min v to reach top = $\sqrt{5gr}$ since $\frac{1}{2}mv^2 - \frac{1}{2}mgr = mg(2r)$ (conservation of energy)

Since sphere fell off between mid-height and top, $\sqrt{2gr} < v < \sqrt{5gr}$.

11 A planet orbits around a star with a period of T and an orbiting radius of r. The planet is then hit by some asteroids, causing its total energy to decrease and its orbital radius to decrease to $0.25r$.

In this new orbit, how many revolutions does the planet make in the same time T?

A 0.13 **B** 0.25 **C** 4.0 **D** 8.0

Ans: **D**

Gravitational force provides centripetal force

$$\frac{GMm}{r^2} = mr\omega^2 = mr(\frac{2\pi}{T})^2$$

$$T^2 = \frac{4\pi^2}{GM}r^3$$

$$T^2 \propto r^3$$

$$\frac{T_1^2}{T_2^2} = \frac{r_1^3}{r_2^3} = \frac{r^3}{(0.25r)^3} = 64$$

$$T_1 = \sqrt{64}T_2 = 8T_2$$

12 Which statement about geostationary orbits on Earth is **false**?

A A geostationary orbit must be directly above the equator.

B All satellites in geostationary orbit must have the same mass.

C The period of a geostationary orbit must be 24 hours.

D There is only one possible radius for a geostationary orbit.

Ans: **B**

Option A: Since orbits can only be centred about the centre of mass of the Earth, only orbits around the equator will allow for geostationary orbit.

Option B: untrue. Mass of satellite doesn't affect the period.

$$\frac{GMm}{r^2} = mr\omega^2 = mr(\frac{2\pi}{T})^2$$

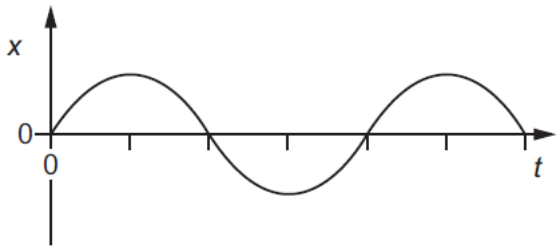
$$T^2 = \frac{4\pi^2}{GM}r^3 \text{ (independent of mass of satellite)}$$

Option C: Since the period of the Earth's rotation is 24 hours, that must also be the period of the orbit.

Option D: Looking at the math in option B answers, $T^2 \propto r^3$. This means that every period corresponds to a specific orbiting radius.

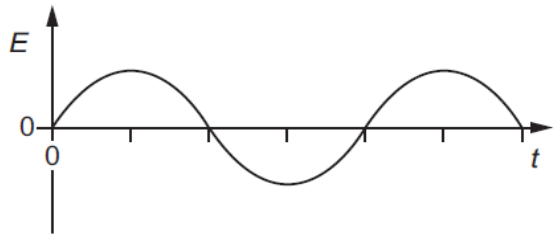
13 A body is undergoing undamped oscillations that are simple harmonic.

A graph of the variation of displacement x of the body with time t is shown.

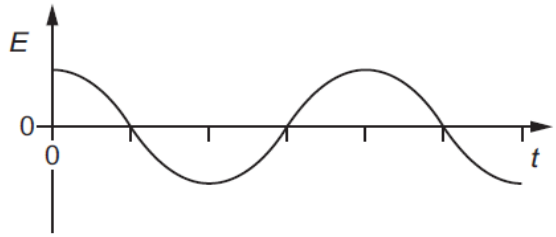


Which graph shows the variation of the kinetic energy E of the body with time t?

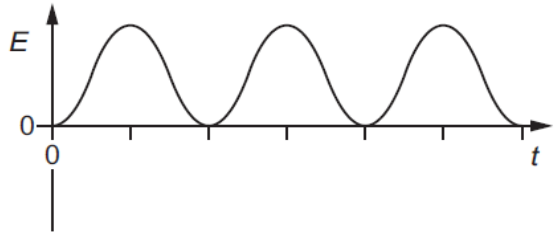
A



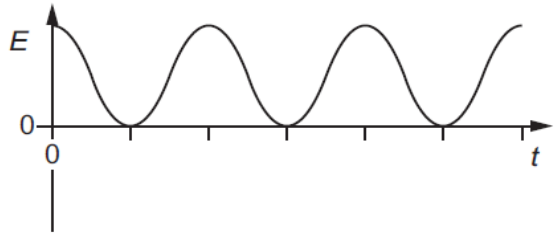
B



C



D

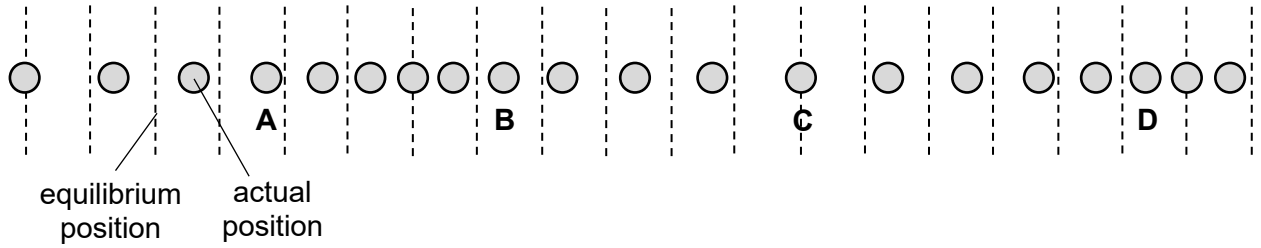


Ans: **D**

At t = 0, displacement of body is x = 0 at equilibrium position which means its velocity and hence KE is maximum. There are 2 cycles in a period. Hence option **D**.

14 A longitudinal sound wave travels from left to right through air.

The diagram below shows the equilibrium positions and the actual positions of a series of air particles through which the sound wave is passing.



Which of the particles has an acceleration that is opposite in direction to its velocity at this instant?

Ans: **D**

Hint: sketch the displacement-distance graph of the particles.

A: The particle is at its amplitude position, so there is no velocity (velocity is zero) at this instant.

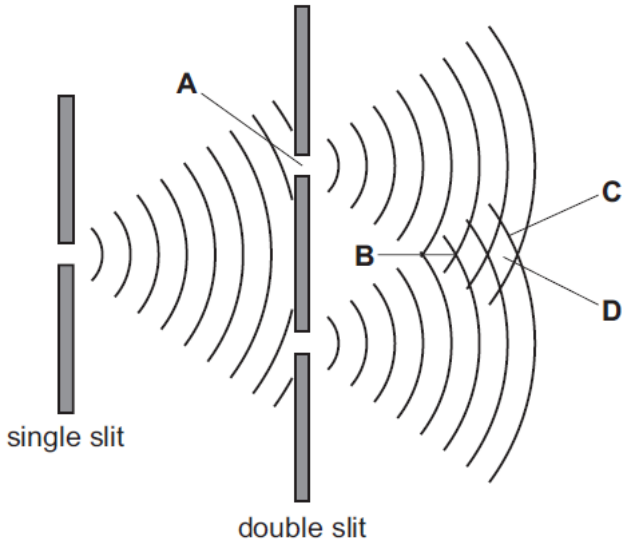
B: The particle is moving towards equilibrium position and velocity is in the same direction as acceleration directed towards equilibrium position.

C: The particle is at the equilibrium position and has maximum velocity but zero acceleration.

D: The particle is moving away from equilibrium position and velocity is in the opposite direction as acceleration directed towards equilibrium position.

15 Light waves from a single slit pass through a double-slit and interfere. The curved lines represent wavefronts.

At which point does complete destructive interference occur?



Ans: **C**

Destructive interference occurs when a crest (curved line) meets a trough (white space).

16 An optical telescope received yellow light of wavelength 6×10^{-7} m from two yellow stars which are 1×10^{11} m apart and can just be resolved by the telescope.

The two stars are 6×10^{16} m from the telescope.

Which statement is **incorrect**?

A The aperture of the telescope has a diameter of about 0.4 m.

B The resolving power of the telescope depends on both the wavelength of light received and the aperture diameter of the telescope.

C The telescope receiving red light of wavelength 7×10^{-7} m can be used to resolve two red stars of the same separation and distance from the telescope as the two yellow stars.

D The telescope receiving yellow light of wavelength 6×10^{-7} m can be used to resolve another pair of yellow stars that are separated by 3×10^{-6} radians.

Ans: **C**

$$s = r\theta, \quad 1\times10^{11} = \left(6\times10^{16}\right)\theta$$

Option A is correct: $\theta = 1.7\times10^{-6}$ rad

$$\theta = \frac{\lambda}{b}, \quad 1.7\times10^{-6} = \frac{6\times10^{-7}}{b}, \quad b = 0.35 \approx 0.4 \text{ m}$$

Option B is correct by Rayleigh's formula.

Option C is incorrect as red light has a longer wavelength, so the telescope's minimum resolvable angle becomes larger than the actual angular separation between the two stars which was at the same distance apart and from the telescope.

Option D is correct as 3×10^{-6} rad is larger than the minimum resolvable angle of 1.7×10^{-6} rad.

17 Which is **not** an assumption of the kinetic theory of gases?

A The molecules move at the same speeds.

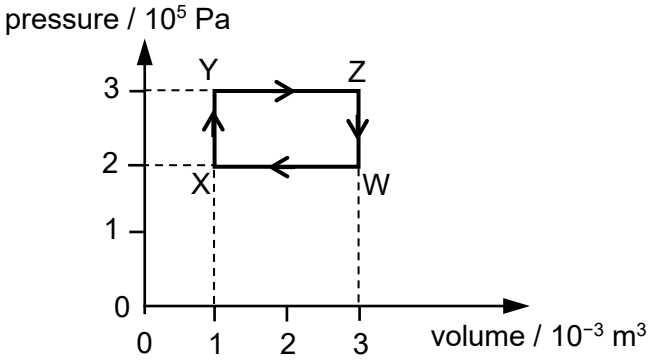
B The motion of the molecules is random.

C The volume of the molecules is negligible compared with the volume of the gas.

D There are negligible forces of attraction or repulsion between molecules except during collisions.

Ans: **A**

18 A fixed mass of ideal gas undergoes the cycle of changes represented by WXYZW in the diagram. There is a decrease in internal energy of 600 J from W to X and an increase in internal energy of 150 J from X to Y.



Which of the following set of information is correct for one complete cycle WXYZW?

	net increase in internal energy / J	net work done on gas / J	net heat absorbed by gas / J
A	0	200	−200
B	0	−200	200
C	450	200	250
D	−450	−200	−250

Ans: **B**

For cyclic process, $\Delta U = 0$

Since its an clockwise cycle, there is net negative work done on the gas

Net work done on gas = area enclosed

$W = -((3-2)\times 10^5)((3-1)\times 10^{-3})$

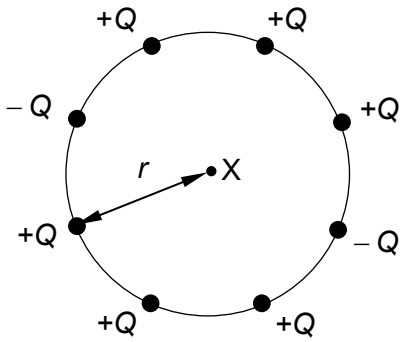
$= -200$ J

$\Delta U = Q + W$

$0 = Q + (-200)$

$Q = 200$ J

19 A point X is surrounded symmetrically by six positive charges +Q and two negative charges −Q at distance r as shown.



How much work is done by the electric field to bring a point charge of +2Q from infinity to position X?

A $-\dfrac{Q^2}{\pi\varepsilon_0 r}$ **B** $-\dfrac{2Q^2}{\pi\varepsilon_0 r}$ **C** $\dfrac{2Q^2}{\pi\varepsilon_0 r}$ **D** $-\dfrac{4Q^2}{\pi\varepsilon_0 r}$

Ans: **B**

Potential is a scalar.

$$V_{centre} = 6\left(\frac{Q}{4\pi\varepsilon_0 r}\right) + 2\left(\frac{-Q}{4\pi\varepsilon_0 r}\right) = \frac{Q}{\pi\varepsilon_0 r}$$

Work done by external force: $U = (+2Q)(V) = (2Q)\left(\dfrac{Q}{\pi\varepsilon_0 r}\right) = \dfrac{2Q^2}{\pi\varepsilon_0 r}$

Hence work done by electric repulsion = $-\dfrac{2Q^2}{\pi\varepsilon_0 r}$

A: forgot 2Q

C: used WD by external force

D: didn't consider sign of charges

20 What is the electrostatic force between a thorium nucleus of charge 1.4×10^{-17} C and an alpha particle when they are separated by a distance of 0.10 pm?

A 4.0×10^{-20} N **B** 4.0×10^{-13} N **C** 4.0 N **D** 8.0 N

Ans: **C**

Potential is a scalar.

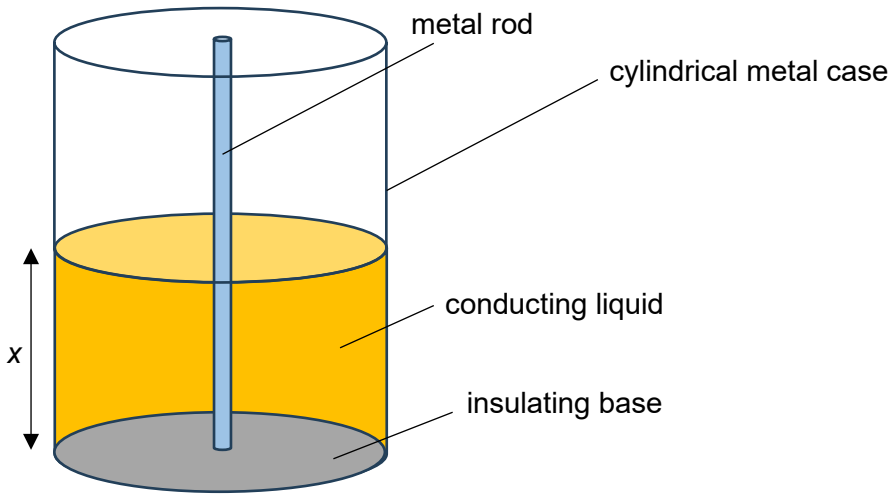
$$F = \frac{Q_{Th}Q_{\alpha}}{4\pi\varepsilon_0 r^2} = \frac{(1.4\times10^{-17})(2\times1.6\times10^{-19})}{4\pi(8.85\times10^{-12})(0.10\times10^{-12})^2} = 4.0 \text{ N}$$

A: forgot to convert pm

B: forgot to divide by r^2

D: took helium charge as 4e.

21 A conducting liquid fills a cylindrical metal case to a depth x as shown below. The case has an insulating base.



The resistance between the case and the metal rod will be

A proportional to x^2

B proportional to x

C inversely proportional to x

D inversely proportional to x^2

Ans: **C**

$$R = \rho\frac{l}{A} = \rho\frac{r}{2\pi r_{average}x} \quad \rightarrow \quad R \propto \frac{1}{x}$$

22 Twenty-five lamps are connected in series to a mains supply. The lamps do not light up when the supply is switched on. A voltmeter gives a zero reading when connected across the third lamp in the set. Which fault cannot be the only one in the circuit?

A The filament of one of the other lamps is broken.

B The filament of the third lamp is broken.

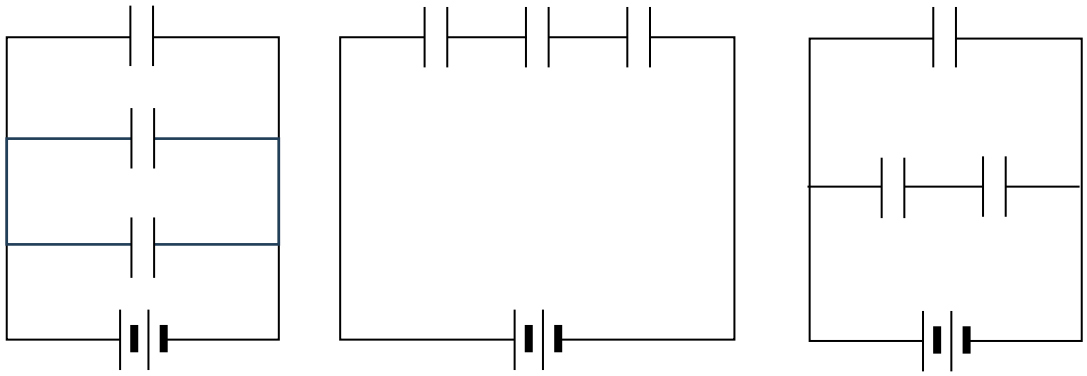
C The fuse in the mains supply has blown.

D There is a break in the wire connecting the set of lamps to the supply.

Ans: **B**

If only B happens, then the pd across the third lamp is the pd across the supply and will not be zero.

23 Three identical capacitors are connected to a battery in three different arrangements as shown below.



Which of the following lists the arrangements in order of increasing capacitance?

A PRQ **B** QRP **C** PQR **D** RQP

Ans: **B**

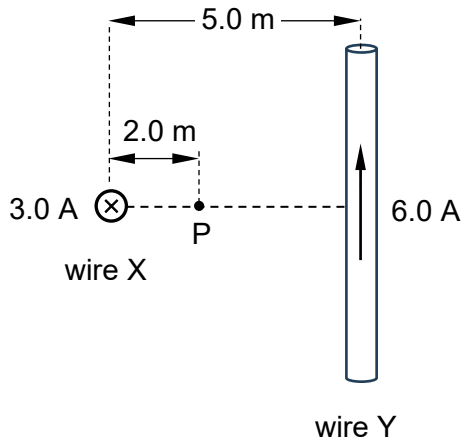
P: $C_{eff} = 3C$

Q: $\frac{1}{C_{eff}} = \frac{3}{C}$

R: $C_{eff} = C + \frac{1}{2/C}$

24 Two long, straight, current-carrying wires, X and Y are at a perpendicular distance 5.0 m apart. The current in X is 3.0 A and into the plane of the paper, while current in Y of 6.0 A is towards the top of the page.

Point P is on the line of shortest distance joining the two wires, at a distance 2.0 m away from wire X as shown.



What is the magnitude of the resultant magnetic flux density at point P?

A 3.5×10^{-12} T

B 5.0×10^{-7} T

C 6.3×10^{-7} T

D 7.0×10^{-7} T

Ans: **B**

Magnetic flux density is a vector.

$$B_x = \frac{\mu_0 I_x}{2\pi d_x} = \frac{(4\pi \times 10^{-7})(3.0)}{2\pi(2.0)} = 3.0 \times 10^{-7} \text{ T (downwards towards bottom of page)}$$

$$B_y = \frac{\mu_0 I_y}{2\pi d_y} = \frac{(4\pi \times 10^{-7})(6.0)}{2\pi(3.0)} = 4.0 \times 10^{-7} \text{ T (out of the page)}$$

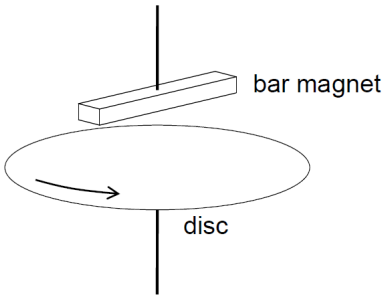
$$B_{\text{resultant}} = \sqrt{B_x^2 + B_y^2} = \sqrt{(3.0 \times 10^{-7})^2 + (4.0 \times 10^{-7})^2}$$
$$= 5.0 \times 10^{-7} \text{ T}$$

A: took epsilon instead.

C: swopped distances

D: took direction addition.

25 A bar magnet is freely suspended so that it hangs horizontally above a flat aluminium disc which is rotating in a horizontal plane, as shown in figure below.



The bar magnet will

A rotate in the opposite direction to the disc.

B rotate in the same direction as the disc.

C remain at rest.

D oscillate to and fro in the horizontal plane.

Ans: **B**

By Lenz's Law, to counter the increase in magnetic flux linkage.

26 A generator produces a current of 60.0 A r.m.s at a voltage of 120 V r.m.s. An ideal transformer steps up the voltage to a peak value of 4500 V and transmits the power through a cable of total resistance 40.0 Ω to the consumers.

What is the percentage of power transmitted to the consumers?

A 94.3 %

B 97.2 %

C 98.6 %

D 99.3 %

Ans: **B**

Primary circuit mean power $<P_p>$ = 120 × 60.0 = 7200 W

For the secondary circuit,

$$I_{rms}\left(\frac{4500}{\sqrt{2}}\right) = 7200$$

$$I_{rms} = \left(\frac{7200\sqrt{2}}{4500}\right)$$

Mean power transmitted P' = 7200 − $I_{rms}^{2}R$

$$= 7200 - \left[\left(\frac{7200\sqrt{2}}{4500}\right)^{2} 40.0\right]$$

Therefore, % power transmitted

$$= \frac{P'}{7200} \times 100\%$$

≈ 97.2 %

27 A green laser pointer and a red laser pointer of the same power cast circular images on a white screen of the same diameter. The wavelengths of the green and red lasers are λ_G and λ_R respectively.



If the pressure exerted by the green laser on the screen is p, what is the pressure exerted by the red laser?

A p **B** $\frac{\lambda_G}{\lambda_R}p$ **C** $\frac{\lambda_R}{\lambda_G}p$ **D** $\left(\frac{\lambda_G}{\lambda_R}\right)^2 p$

Ans: **A**

Momentum of photon = $\frac{h}{\lambda}$, Energy of photon = $\frac{hc}{\lambda}$

Rate of emission of photons = $\frac{P}{hc/\lambda}$ where P is power

Assuming all photons are reflected,

Force exerted on photons = $\frac{P}{hc/\lambda}\left(\frac{h}{\lambda}-\left(-\frac{h}{\lambda}\right)\right)=\frac{2P}{c}$ (N2L)

= Force on screen (N3L)

Pressure = $\frac{\frac{2P}{c}}{\pi\frac{D^2}{4}}=\frac{8P}{c\pi D^2}$ where D is diameter of image

Since pressure is independent of λ, and both lasers have the same power and produce images of the same diameter, pressure by red laser is also p.

28 Which statement best describes the 'electron cloud' in an atom?

A A shell of negative charge at a fixed radius.

B A region where the electron's charge is uniformly distributed.

C The path the electron follows around the nucleus.

D A region where the probability of finding the electron is high.

Ans: **D**

The state of the electron of the atom is represented as a wavefunction ψ and the square of the wavefunction amplitude $|\psi|^2$ is the probability density function.

An electron cloud is the visual representation of the probability density function which dictates where an electron is likely to be found around the nucleus of the atom.

29 The ratio of ^{14}C : ^{12}C in living things is fixed when they are alive but decreases after death because of the radioactive decay of ^{14}C. ^{14}C has a half-life of 5730 years. A 10.0 g sample of wood cut from a live tree is measured to have an activity of 120 min^{-1}.

If a 1.0 g sample of an old wooden relic is found to have an activity of 8.05 min^{-1}, how long ago was this wooden relic cut from a live tree?

A 1470 years **B** 3300 years **C** 9950 years **D** 22300 years

Ans: **B**

Since $A \propto N$,

$$A_{1.0g \text{ live wood}} = \frac{120}{10} = 12.0 \text{ min}^{-1}$$

$$A = A_0 e^{-\lambda t}$$

$$8.05 = 12.0e^{-\frac{\ln 2}{5730}t}$$

$$t = 3300 \text{ years}$$

30 Uranium-235 undergoes fission in a reaction shown below, releasing 195 MeV of energy. The binding energy per nucleon for uranium-235 is 7.6 MeV. The binding energy per nucleon for caesium and rubidium are each approximately equal to **X** MeV.

$$^{235}_{92}U + ^{1}_{0}n \rightarrow ^{143}_{55}Cs + ^{90}_{37}Rb + 3\,^{1}_{0}n$$

What is the value of **X**?

A 6.7 **B** 6.8 **C** 8.4 **D** 8.5

Ans: **D**

$$\text{Energy released} = BE_{products} - BE_{reactants}$$

$$195 = X(143) + X(90) - 7.6(235)$$

$$X = 8.50 \text{ MeV}$$

End of Paper